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Insights into acidic deep eutectic solvent-mediated softwood pretreatment for high fermentable sugar yield and efficient lignin recovery

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ABSTRACT

This study addresses the critical challenges of softwood pretreatment, focusing on Chinese fir as a model substrate to tackle its high lignin content and recalcitrant structure. To enhance the fractionation efficiency of softwood components and improve enzymatic glucose yield, four deep eutectic solvent (DES) systems—ChCl/LA, ChCl/FA, TEBAC/LA, and TEBAC/FA—were systematically evaluated. Using density functional theory (DFT) calculations and molecular dynamics (MD) simulations, TEBAC/LA (1:4) was identified as the optimal DES system, exhibiting exceptional binding energy (-162.37 kcal/mol) and strong hydrogen bonding interactions that effectively disrupted lignin's hydrogen bond network. Furthermore, Pearson correlation analysis between the physicochemical properties of DES (e.g., density, viscosity, surface tension, and Kamlet-Taft parameters) and enzymatic hydrolysis yield revealed that low surface tension, high Kamlet-Taft α, and high π parameters significantly enhance pretreatment efficiency. Guided by these computational insights, experimental validation revealed that TEBAC/LA achieved outstanding lignin removal (97.41 %) and glucose yield (97.09 %) under optimized conditions (150 °C, 90 min). This work not only advances the understanding of DES-biomass interactions but also demonstrates the power of integrating molecular simulations with experimental validation to design efficient biomass pretreatment systems. This study provides a scalable and sustainable approach for optimizing solvent systems, advancing renewable energy and material production.

1. Introduction

Softwood, a prevalent form of woody biomass found in the Northern Hemisphere, presents significant challenges owing to its elevated contents of cellulose and lignin (Hossain et al., 2019). Previous studies have shown that damage to softwood structures requires more severe pretreatment strengths (Ouyang et al., 2021; Nitsos et al., 2017; Ramos et al., 2003). For most biomasses, common industrial acid pretreatment at temperatures between 80 and 150 °C for 0.5–3 hours could remove over 70 % of hemicellulose and over 60 % of lignin, achieving enzymatic yields of over 85 % (Liu et al., 2022). However, when pretreating softwoods, Nguyen et al. even increased the temperature to 200–230 °C during the acid pretreatment, only a 40 % enzymatic yield was achieved (Nguyen et al., 2000). By comparison, combined pretreatment methods seem to obtain the desired enzymatic yield (Rabelo et al., 2023). Formic acid-assisted ethanol at 235 °C was used to pretreat spruce and the enzymatic yield approached 100 %, but there was significant cellulose loss (Agnihotri et al., 2015). Therefore, recent attention has shifted to ionic liquids pretreatment. Gschwend et al. used N, N-dimethylbutylammonium hydrogen sulfate for pine pretreatment at 170 °C, releasing 464 mg of glucose per gram of pine while also separating lignin (Gschwend et al., 2019). Its good performance could imply that pretreatment with solvents rich in hydrogen bonds merits further exploration and development.

Deep eutectic solvents (DES) are a type of solvent that is abundant in hydrogen bonds, and they are regarded as one of the most promising options for the pretreatment of biomass (Zhang et al., 2022a). In the process of biomass pretreatment, DES utilizes a gentle acid-base catalytic method to specifically break unstable ether bonds that link phenylpropane units. This results in the depolymerization of lignin and its separation from the biomass, all while maintaining the integrity of cellulose and avoiding its degradation (Alvarez-Vasco et al., 2016; Shen et al., 2019). DES is now widely used in the pretreatment of grassy and hardwood biomasses, not only achieving high enzymatic yields but also

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allowing for the selective extraction of high-purity lignin (Li et al., 2022). However, the use of DES in softwood biomass is still limited, with challenges such as high solvent consumption, extended pretreatment durations, and high temperatures. These issues highlight the need for new DES systems (Wu et al., 2023; Gholami et al., 2024). Choline chloride (ChCl) is the commonly used HBA in DES pretreatment, which not only can be mixed with various HBDs, such as amino acids, carboxylic acids, alcohols p-toluenesulfonic acid, etc., but also removes lignin with remarkable effect (Liu et al., 2019). Triethylbenzylammonium chloride (TEBAC) has more benzene ring structure compared to ChCl, and as a cationic surfactant, it has a higher phase-transfer catalytic ability than ChCl, and has a more obvious effect of delignification (Liu et al., 2021). Formic acid (FA) and lactic acid (LA) as organic acids, it has been found that DES pretreatment of Gramineae with FA and LA as HBD was effective, not only separating more than 65 % of hemicellulose and lignin and achieving favorable enzymatic hydrolysis (over 75 %), but also obtaining lignin purity of more than 85 % (Kang et al., 2023; Zhang et al., 2022b). Additionally, the mechanisms of action of DES in the separation of lignocellulosic components remain underexplored. With the advancement of computational chemistry, molecular dynamics (MD) can be used to study the interaction mechanisms between DES and lignocellulosic components (Xu et al., 2021; Wang et al., 2022a), while density functional theory (DFT) calculations offer microscopic insights into the weak interactions among these components, providing theoretical support for MD studies (Wang et al., 2022b).

This study aimed to identify an effective one-step DES pretreatment system capable of efficiently breaking down the enzymatic barriers in *Chinese fir* residue. The selection of the DES system was based on the comparison of the physicochemical properties of DES hydrogen donors and acceptors, analyzing their relationship with pretreatment outcomes, and exploring interactions between the DES system and lignin model compounds using molecular dynamics (MD) simulations and density functional theory (DFT) calculations. The pretreatment system was optimized to develop a method capable of efficiently separating softwood biomass lignin and overcoming enzymatic barriers, offering a potential pretreatment technology for the biorefining of softwood biomasses.

2. Materials and methods

2.1. Materials

The raw material for pretreatment in this experiment was *Chinese fir* residue from Linyi, Shandong Province. Before use, the wood residue underwent sieving to achieve a particle size ranging from 20 to 80 mesh. Cellulase (Cellic® CTec2) used for enzymatic hydrolysis was purchased from Sigma-Aldrich. Other chemical reagents were sourced from Sinopharm Chemical Reagent Co., Ltd. (Shanghai, China) and Shanghai Aladdin Biochemical Technology Co., Ltd.

2.2. Preparation and pretreatment of DES

The hydrogen bond acceptor (HBA) and hydrogen bond donor (HBD) were combined in different molar ratios (1:2/1:4) in a dry glass vial to synthesize the DES. The mixture was then heated to 60 °C until a uniform solution formed.

5 g of *Chinese fir* residue was added to a 100 mL Polytetrafluoroethylene reaction vessel, and the prepared DES solvent was added at a solid-to-liquid ratio of 1:10 (Solid/Liquid ratio of 1:10 g/g). The mixture was then pretreated at different temperatures (130–160 °C) for varying durations (15–90 min). Following pretreatment, the mixture was cooled, and solid-liquid separation was performed to isolate the solid residues and liquid. The solid residue was then washed with 95 % ethanol and a 1 mol/L NaOH solution, followed by rinsing with distilled water until a neutral pH was achieved. Lignin was precipitated from the liquid portion by adding ten times its volume of water. The precipitated lignin was then washed with distilled water until neutral, followed by vacuum drying and weighing. All experiments were repeated twice.

2.3. Enzymatic hydrolysis

The concentration of the substrate for hydrolysis was set at 5 % (weight/ volume), and the amount of cellulase added was 20 FPU/g of glucan (Qiao et al., 2023; Ma et al., 2024). The enzymatic reaction took place at a temperature of 50 °C for 48 hours. Samples were periodically collected during enzymatic hydrolysis, and then appropriately diluted and analyzed for glucose concentration using High-Performance Liquid Chromatography (HPLC). All experiments were repeated twice. The yield of glucose from the enzymatic hydrolysis was determined using the following calculation:

$$Enzymatic\ yield\% = \frac{Glucose\ in\ enzymatic\ hydrolysis \times 0.9}{Initial\ glucan\ concentration} \times 100\% \quad (1)$$

2.4. Analytical methods

2.4.1. Characterization of DES physicochemical properties

The density of DES was calculated according to Shafie et al. (2019).

The chemical functional groups of DES were characterized using Fourier Transform Infrared Spectroscopy (FT-IR) (Shimadzu IRTracer-100, Japan) in ATR mode, covering a scanning range from 400 to 4000 cm^{-1} .

The apparent viscosity of DES was measured using a MARS60 micro-infrared combined rheometer (USA). Around 0.2 mL of the DES sample was applied to the lower plate, and the cone geometry was adjusted to the truncation gap and allowed to stabilize. The temperature of the lower plate was regulated using a cone-plate temperature control system, and stable shear viscosity measurements were conducted at 25 °C and 150 °C with a shear rate of 500 s^{-1} .

The surface tension of DES solutions at 25 °C and 150 °C was determined using a video-based optical contact angle tensiometer (Dataphysics-OCA20) by the pendant drop method. The accompanying software (SCA 20) captured the shape of the liquid drop, and the surface tension of DES was automatically computed using the Young-Laplace equation.

2.4.2. Determination of solvent color development parameters of DES

For solvent color development parameters (π*, β and α) of DES, 4-nitroanisole, 4-nitroaniline, and Nile red were used as dye indicators for determination, respectively (Florindo et al., 2018). Absorption spectra of mixed solutions of dye and DES were obtained by scanning with a UV-Vis spectrophotometer. The parameters for solvent color development were calculated according to Liang et al. (2021).

2.4.3. Quantum chemical calculations and molecular dynamics simulations

Molecular dynamics (MD) simulations were carried out in MS to optimize the complex and calculate the hydrogen bond distances between HBA and HBD (Agrafiotis et al., 1996). Density functional theory (DFT) calculations were performed using the Gaussian 16 software (Frisch and Trucks, 2016). Veratrylglycerol-b-guaiacyl ether (VG) was considered an appropriate model compound for examining the degradation of lignin during pretreatment, as it possessed functional groups and bonds akin to those found in lignocellulosic biomass lignin, including *β*-O-4 ether bonds and phenolic hydroxyl groups (Ji and Lv, 2020). The structures of four monomers (VG, ChCl, TEBAC, LA) were optimized using the B3LYP-D3(BJ)/def2-SVP level of theory (Stephens et al., 1994; Grimme et al., 2010). Single-point energy calculations were performed at the M062x-D3/6–311 G(d,p) level to obtain molecular wave function information (Zhao et al., 2008). Electrostatic potential (ESP) was performed in Multiwfn8 package (Lu and Chen, 2012). All atoms were parameterized using the GAFF (Wang et al., 2004) force field and Auxiliary Tools of Force Field (AuToFF). The interactions between

VG and different molecules were studied through MD simulations. The MD simulations were conducted using the GROMACS 2021 software package (Van Der Spoel et al., 2005). A 10 × 10 × 10 nm box was constructed, and 60 VG monomers were randomly added to the box along with 800 HBD and 200 HBA molecules in proportion. Before MD calculations, the initial energy of each system was optimized using the steepest descent method with a force tolerance of 1 kJ/(mol·nm) and a maximum step size of 0.002 ps. Periodic boundary conditions were applied in all three dimensions. The MD simulations were conducted in an NPT ensemble for 50 ns, maintaining a pressure of 1 bar and a temperature of 423.15 K (150 °C). Electrostatic interactions were assessed using the particle-mesh Ewald (PME) method with fourth-order interpolation, while short-range van der Waals interactions were calculated with a cutoff distance of 1.0 nm (Darden et al., 1993).

2.4.4. Analysis of chemical composition of solid samples

The samples' chemical composition was examined following the protocol established by the National Renewable Energy Laboratory (NREL), which included determining the structural carbohydrates and lignin content of all solid samples (Sluiter, 2008). Monosaccharides obtained from sulfuric acid hydrolysis were quantified using high-performance liquid chromatography (HPLC) and ion chromatography (IC). All experiments were repeated twice. The solid yield, recovery rate, and corresponding removal rate (%) of each component were calculated using the following formulas (Lan et al., 2020).

$$\textit{Solid recovery}\ (\%) = \frac{\textit{Mass of pretreated Chinese fir residue}}{\textit{Mass of raw Chinese fir residue}} \times 100\% \quad (2)$$

$$\textit{Component recovery}\ (\%) = \frac{\textit{Component content in pretreated residue} \times \textit{Solid recovery}}{\textit{Total component in unpretreated Chinese fir residue}} \times 100\% \quad (3)$$

$$\textit{Component}\ \text{Removal}\ (\%) = 100\% - \textit{component recovery}\ (\%) \quad (4)$$

2.4.5. Other analytical methods

For the soluble sugar content of the samples, high-performance liquid chromatography (HPLC, Agilent 1260) was employed. The conditions for this analysis included an HPX-87H column (7.8 mm × 300 mm) at 55 °C, a mobile phase of 5 mM H_2SO_4, a flow rate of 0.6 mL/min, and a refractive index detector.

For the other monosaccharides (mannose, xylose, galactose and arabinose) in the sample were simultaneously measured using high-performance anion-exchange chromatography (Dionex ICS-5000). The chromatography settings included a CarboPac PA 10 column (4 mm × 250 mm) at a temperature of 30 °C, with a mobile phase of 3 mmol/L NaOH and 200 mmol/L NaOH, and a flow rate of 0.2 mL/min.

To assess the linear relationships between various factors, Pearson correlation analysis was conducted, and correlation heatmaps were created using GraphPad Prism 9.0 software.

3. Results and discussion

3.1. Formation and physicochemical properties of deep eutectic solvents

In this study, choline chloride (ChCl) and triethylbenzylammonium chloride (TEBAC) were chosen as HBAs, while formic acid (FA) and lactic acid (LA) were selected as HBDs. ChCl and LA are commonly used HBA and HBD, respectively, both having two hydrogen bond donor/acceptor sites. TEBAC, compared to ChCl, contained a benzene ring structure. Although it had only one hydrogen bond acceptor site, the benzene ring structure facilitated π-π stacking interactions with the aromatic rings in lignin. Although FA had only one hydrogen bond donor site, it was more acidic. The morphology of the synthesized DES solutions at room temperature was shown in Fig. S1, where the color of the ChCl-based DES solution was transparent, while the color of the TEBAC-based DES solution showed a light yellow color.

3.1.1. FTIR analysis

Fourier Transform Infrared (FT-IR) spectroscopy was widely used to study molecular interactions, as it could characterize samples by analyzing the presence of functional groups within atomic bonds or molecules. Therefore, infrared detection was carried out on these eight DES solutions. Earlier research has indicated that in DES solutions (AlOmar et al., 2016), the O-H stretching band was found between 3600 and 3200 cm^{-1}, whereas the N-H stretching band was located between 3400 and 3000 cm^{-1}, which overlaps with the O-H stretch. Additionally, the stretching of saturated ester bonds occurred in the range of 1300–1100 cm^{-1}.

As seen in Fig. 1, the spectral analysis of the ChCl/TEBAC series revealed that the O-H stretching band for ChCl was situated at 3300 cm^{-1}, while the CH_2 bending vibration was observed at 1470 cm^{-1}. C-N stretching bands were detected at 1376 cm^{-1} and 892 cm^{-1} (Shafie et al., 2019). For TEBAC, the O-H stretching band was also found at 3300 cm^{-1}, with the bending vibration of CH_2 appearing at 1450 cm^{-1}, and the C-N stretching bands located at 1370 cm^{-1} and 900 cm^{-1}. Absorption peaks characteristic of the benzene ring were noted at 750 and 690 cm^{-1}. In the LA/FA series spectra, the O-H stretching band of the carboxyl group in LA was located at 2990 cm^{-1}, the C═O stretching band at 1720 cm^{-1}, and the C-O stretching band at 1200 cm^{-1}. For FA, the O-H stretching band of the carboxyl group was located at 2920 cm^{-1}, with the C═O stretching band at 1720 cm^{-1} and the C-O stretching band at 1190 cm^{-1} (Gautam et al., (2020). Notably, the bands at 1376, 892, and 600 cm^{-1} for ChCl exhibited broadening upon the addition of LA/FA, suggesting the formation of hydrogen bonds among the components of DES (Shafie et al., 2019). Similarly, the spectral analysis of TEBAC revealed that the absorption bands at 1370 and 900 cm^{-1} exhibited broadening upon the incorporation of LA/FA, which implied the formation of hydrogen bonds among the components of DES. The broadening of the O-H bands and the wavenumber shift of the C-O bands in the LA/FA spectra indicated hydrogen bonding and proton transfer between LA/FA and ChCl/TEBAC (Li et al., 2020). Furthermore, as the molar ratio increased, the O-H stretching band of ChCl/TEBAC shifted to higher wavenumbers, suggesting that more Cl^- ions in ChCl/TEBAC induce stronger hydrogen bonding interactions with the hydroxyl groups in LA/FA, resulting in the formation of OH···Cl bonds (Duque et al., 2023).

3.1.2. Molecular modeling of interaction relationships

Section 3.1.1 utilized Fourier Transform Infrared (FT-IR) to confirm that all eight types of DES were capable of forming hydrogen bonds. Understanding the interactions between different DES required examining the characteristics of hydrogen bonding within these solvents (Ramírez et al., 2018). This analysis involved using MS to explore hydrogen donor-acceptor interactions and to optimize molecular structures across these eight DES, with the results presented in Fig. 2. Both LA and FA effectively paired with ChCl and TEBAC. The potential hydrogen bond distances between hydrogen donors and acceptors in the eight DES

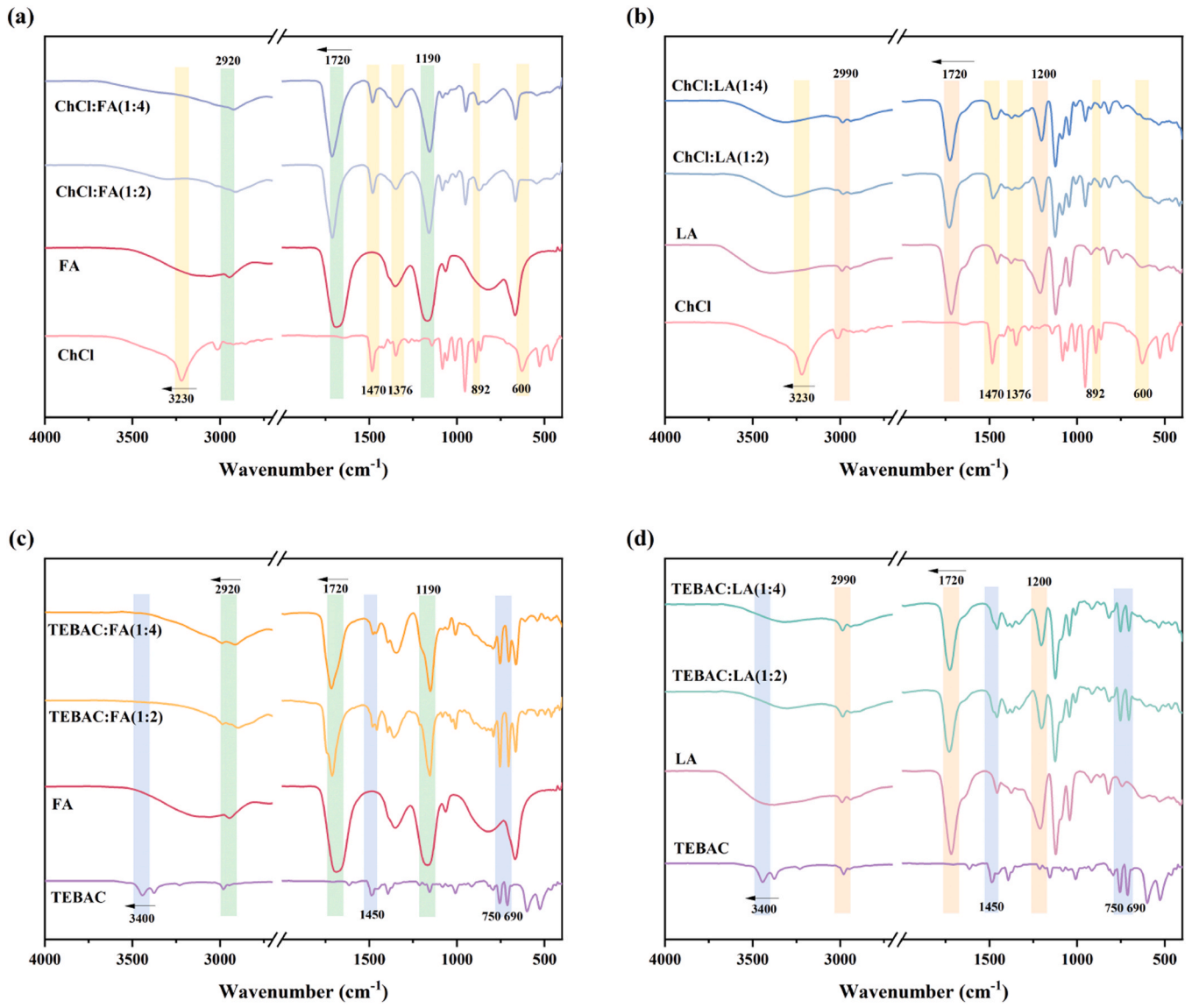


Fig. 1. FTIR spectra for various DES solutions are presented. (a) FTIR spectra for pure ChCl, pure FA, and ChCl/FA at various molar ratios. (b) FTIR spectra for pure ChCl, pure LA, and ChCl/LA at different molar ratios. (c) FTIR spectra for pure TEBAC, pure FA, and TEBAC/FA at various molar ratios. (d) FTIR spectra for pure TEBAC, pure LA, and TEBAC/LA at different molar ratios.

were also calculated, with the findings listed in Table S1. In TEBAC/LA or FA DES, the hydrogen atom in the LA/FA carboxyl group could form an OH···Cl anionic hydrogen bond with Cl^- in TEBAC. It was observed that the anionic hydrogen bonding distance in TEBAC /LA (3.375 Å) was shorter than that in TEBAC/FA (3.670 Å). DES systems based on ChCl/LA or FA showed similar trends. Through this configuration optimization analysis, the pivotal role of hydrogen bonding in the interactions within DES was identified. LA and FA were able to effectively interact with Cl^- in both TEBAC, forming stable anionic hydrogen bonds that led to charge delocalization. As hydrogen bond distances decreased, the attractive forces between the DES increased, which encouraged more hydrogen protons to move towards Cl^-, resulting in further proton delocalization and an increase in the solvent's acidity (Xia et al., 2018).

3.1.3. *Physical and chemical properties of DES*

The data in Table 1 showed that the density of ChCl/LA was approximately 1.2 g/cm^3, while ChCl/FA had a density of about 1.18 g/cm^3 . TEBAC/LA exhibited a density of around 1.1 g/cm^3 , and TEBAC/FA had the highest density at about 1.3 g/cm^3 . Hydroxyl groups in LA/FA and Cl^- in ChCl/TEBAC formed complexes of various sizes via hydrogen bonds. This interaction reduced the free space between the components of the deep eutectic solvent (DES), leading to variations in molecular packing and consequent changes in density (García et al., 2015). Literature indicated that as the relative proportion of HBD in the DES components increased, the density also increased (Van Osch et al., 2017), our findings were consistent with this observation.

Generally, DES solutions showed high viscosity because of a dense network of hydrogen bonds among the components, which limited the mobility of free substances (Abbott, 2004). From Table 1, it could be observed that the TEBAC/LA series exhibited higher viscosity at room temperature, indicating strong hydrogen bond interactions between TEBAC and LA. The viscosity of most DES decreased as the temperature rose and the amount of HBA decreased (Haedi et al., 2017), and when the temperature rose to 150 °C, the viscosity of the selected eight types of DES dropped to below 10 mPa·s. Research by Kuldeep Singh et al. has shown that with rising temperatures, the viscosity of TEBAC-based DES decreased (Singh et al., 2023). Lower viscosity implied better fluidity of the solution, facilitating contact with lignocellulose and aiding in the separation of lignin.

Measuring surface tension was crucial for comprehending the energy

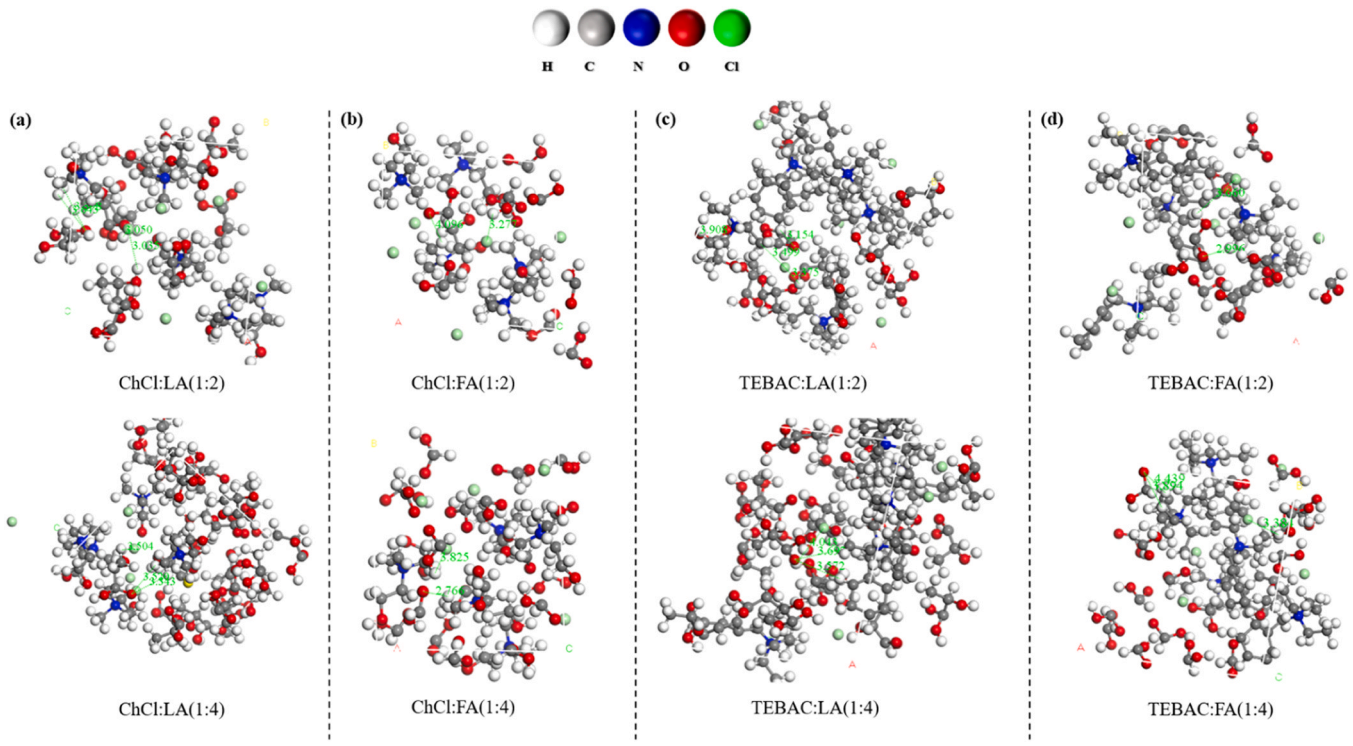


Fig. 2. Structures of optimised DES complexes formed by different HBAs and HBDs. (a) Structure of the ChCl/LA series of DES. (b) Structure of ChCl/FA series DES. (c) Structure of TEBAC/LA series DES. (d) Structure of TEBAC/FA series DES.

Table 1
Physicochemical properties of different DES solvents.

	Density (g/cm^3)	π*	β	α	25 ℃		150 ℃	
					Viscosity (mPa·s)	Surface tension (mN/m)	Viscosity (mPa·s)	Surface tension (mN/m)
ChCl+LA (1:2)	1.20	1.10	0.54	0.98	163.10	47.70	9.19	29.20
ChCl+LA (1:4)	1.22	1.20	0.35	1.04	38.68	47.02	7.52	31.75
ChCl+FA (1:2)	1.17	1.06	0.19	1.30	23.39	69.26	7.17	53.21
ChCl+FA (1:4)	1.18	1.16	0.11	1.35	9.60	66.89	7.30	50.41
TEBAC+LA (1:2)	1.11	1.08	0.60	0.78	927.75	55.54	4.42	15.93
TEBAC+LA (1:4)	1.14	1.15	0.46	0.94	262.05	45.57	5.57	18.96
TEBAC+FA (1:2)	1.29	0.82	0.77	0.71	327.16	69.11	1.88	55.58
TEBAC+FA (1:4)	1.31	0.92	0.59	0.82	10.31	54.32	2.05	44.14

needed to expand the surface area of liquids and the intermolecular interactions between hydrogen donors and acceptors in DES. The surface tension of DES was affected by various factors, including intermolecular forces, temperature, molar ratios, and the characteristics of HBA/HBD (Omar and Sadeghi, 2022). Table 1 clearly indicated that at room temperature, the surface tension of TEBAC-based DES was typically greater than that of ChCl-based DES. This indicated more extensive hydrogen bonding within TEBAC/LA, leading to stronger cohesion among molecules, thus requiring greater intermolecular attraction at the liquid surface to counteract external forces and create surface molecules to prevent surface disruption (Shafie et al., 2019). At elevated temperatures, the reduction in surface tension was mainly caused by the weakening of intermolecular forces.

Previous research has shown that Kamlet-Taft solvatochromic parameters could elucidate the relationship between the polarity of DES and their effectiveness in extracting lignin from biomass (Xie et al., 2023). The parameters α, β, and π* correspond to the acidity of hydrogen bonds, the basicity of hydrogen bonds, and the dipolarity of DES. According to data from Table 1, an increase in molar ratio resulted in a trend of increasing π values, decreasing β values, and increasing α values for the same type of solvent. It was also observed that the fluctuations in these parameter values correlated with HBD. The π parameter was used to assess the solvent's ability to stabilize charges or dipoles; the higher the π value, the greater its stabilizing capacity. Studies have shown that the acidity of hydrogen bonds (α) in DES influences the breakdown of lignin-carbohydrate complexes (LCC) found in biomass (Liang et al., 2021). A rise in the α value improved the effectiveness of lignin extraction from biomass. Based on the foregoing analysis, it was

hypothesized that ChCl/LA and TEBAC/LA possessed strong pretreatment performance.

3.2. The pretreatment effect of different DES reagents for the Chinese fir residue

3.2.1. Effect of different binary DES pretreatments on Chinese fir residue

The physicochemical properties of DES could provide a detailed description of its performance, allowing for an initial assessment of its pretreatment effectiveness. Therefore, based on the results of the previous experiments, the four groups of DES with relatively strong performance were chosen to carry out the pretreatment of *Chinese fir* residue. Initially, the component content of the raw material was measured, with results shown in Table S2 indicating cellulose content at 40.67 %, hemicellulose content at 14.17 %, and lignin content at 30.73 %, which were close to the reported contents of these major components in softwood (Ouyang et al., 2019). Fig. 3a illustrated the alterations in chemical composition and quality of the samples following pretreatment. For the ChCl/TEBAC-based DES, when LA was the HBD, the glucan content proportion and retention were higher compared to when FA was the HBD, with glucan retention exceeding 90 % and lignin removal rates similar, both around 95 %. However, with FA as the HBD, the solid recovery rate was less than 40 %, and the glucan recovery rate was around 80 %, resulting in greater losses. As illustrated in Fig. 3b, the enzymatic hydrolysis yields of pretreated *Chinese fir* residue utilizing ChCl+LA (1:4), TEBAC+LA (1:4), and TEBAC+FA (1:4) all exceeded 95 %. In contrast, the direct enzymatic hydrolysis yield of the raw material was only 8.15 %, demonstrating the excellent pretreatment effect. However, the enzymatic hydrolysis yield of *Chinese fir* residue pretreated with ChCl+FA (1:4) was only 44.94 %, significantly lower than expected, suggesting that the high temperatures may have caused partial dehydration of glucan and conversion of xylose into inhibitors in the *Chinese fir* residue (Zhang et al., 2022c), affecting the binding of cellulase to the material.

3.2.2. DFT calculations of DES interactions with lignin systems

Two groups of ChCl+LA (1:4) and TEBAC+LA (1:4) were selected as the two groups with better pretreatment effect, and the weak interaction of these two groups of DES on lignin was investigated by DTF calculations, so as to select the most ideal combination of pretreatment. Fig. 4a displayed the optimized geometric structures and electrostatic potential surfaces of the monomers LA, ChCl, TEBAC, and VG. The blue regions indicated a negative electrostatic potential, the white regions indicated a neutral potential, and the red regions indicated a positive potential. The hydroxyl hydrogen in VG displayed a positive electrostatic potential, whereas the phenyl ring adjacent to the β-O-4 ether bond exhibited a negative potential. Similarly, the Cl^- in ChCl and TEBAC also had a negative electrostatic potential. Therefore, the negatively charged Cl^- could readily form hydrogen bonds with the positively charged hydroxyl groups of VG, facilitating the separation of lignin and hemicellulose from the biomass surface (Tang et al., 2024). Fig. 4b illustrated the most stable structures of the optimized VG-ChCl-LA and VG-TEBAC-LA systems, where the main interaction between VG and DES was hydrogen bonding. Compared to VG-ChCl-LA (-159.54 kcal/mol), the interaction energy of the VG-TEBAC-LA system was greater (-162.37 kcal/mol), indicating stronger molecular interactions in VG-TEBAC-LA. This indicated that TEBAC-LA interfered with the hydrogen bonding present in the original lignin structure by establishing robust hydrogen bonding interactions with the lignin itself (Wang et al., 2022b).

The Reduced Density Gradient (RDG) method was employed to examine the weak interactions within the DES-lignin system, aiming to gain deeper insights into how these two DES systems influence lignin. The isosurface of the RDG function revealed regions of weak interactions, including van der Waals forces, hydrogen bonding, and site resistance repulsion. Fig. 4c and d showed the RDG analysis and its scatter plots, where the green areas represented weak interactions, blue areas indicated hydrogen bonding, and red areas represented potential resistive effects. It could be observed that the hydrogen bonding interactions in both systems were concentrated between the anions and the hydrogen bond donors. Compared to the VG-ChCl-LA system, the scatter plot for VG-TEBAC-LA had a denser concentration of blue areas, indicating that the interaction strength between TEBAC/LA and VG, mediated by hydrogen bonding, was more pronounced (Wang et al., 2023). In addition, the interaction energies of the two systems with VG at the pretreatment temperature (150 °C) were calculated as in Table S3. The overall interaction energies of both systems appeared to be comparable; however, the interaction energy between TEBAC and VG (-4561.77 kcal/mol) was higher than that between ChCl and VG (-3849.88 kcal/mol). This indicated that TEBAC was more influential than ChCl in the lignin separation process within DES, supporting the previously mentioned findings.

3.2.3. Factors impacting cellulase digestibility

To examine how substrate-related factors and the physicochemical

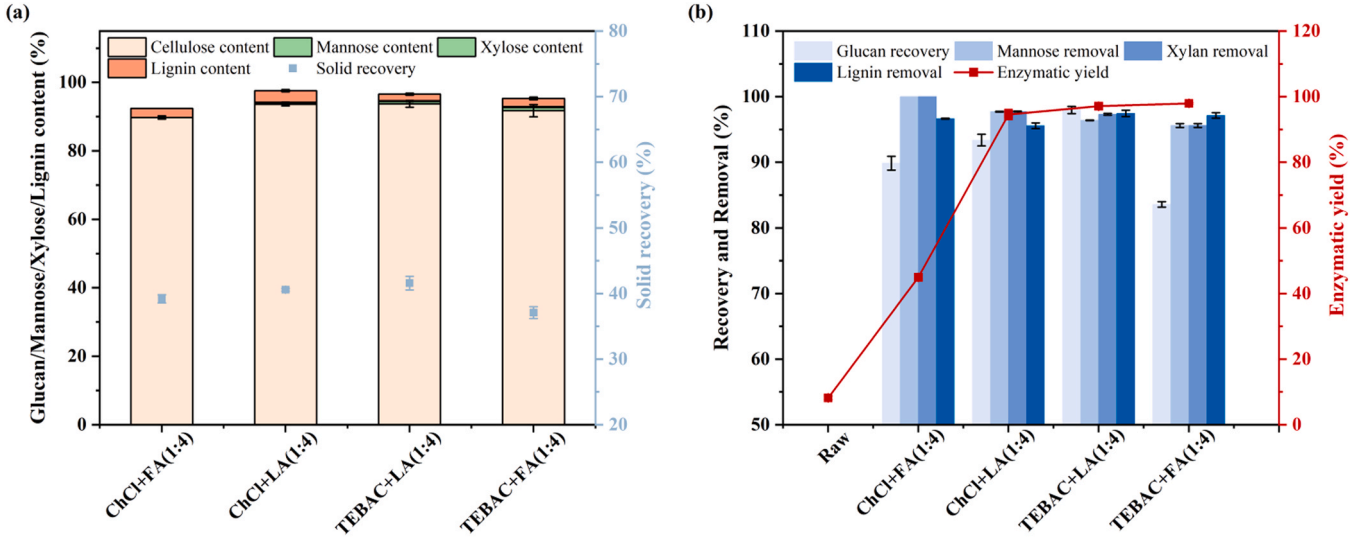


Fig. 3. Pretreatment and enzymatic effects of different DES systems on *Chinese fir* residue. (Reaction conditions: temperature of 150 °C and time of 90 min.) (a) Content and solids recovery of each component following pretreatment with various DES systems. (b) Retention, removal and 48 h enzymatic yield of each component following pretreatment with various DES systems.

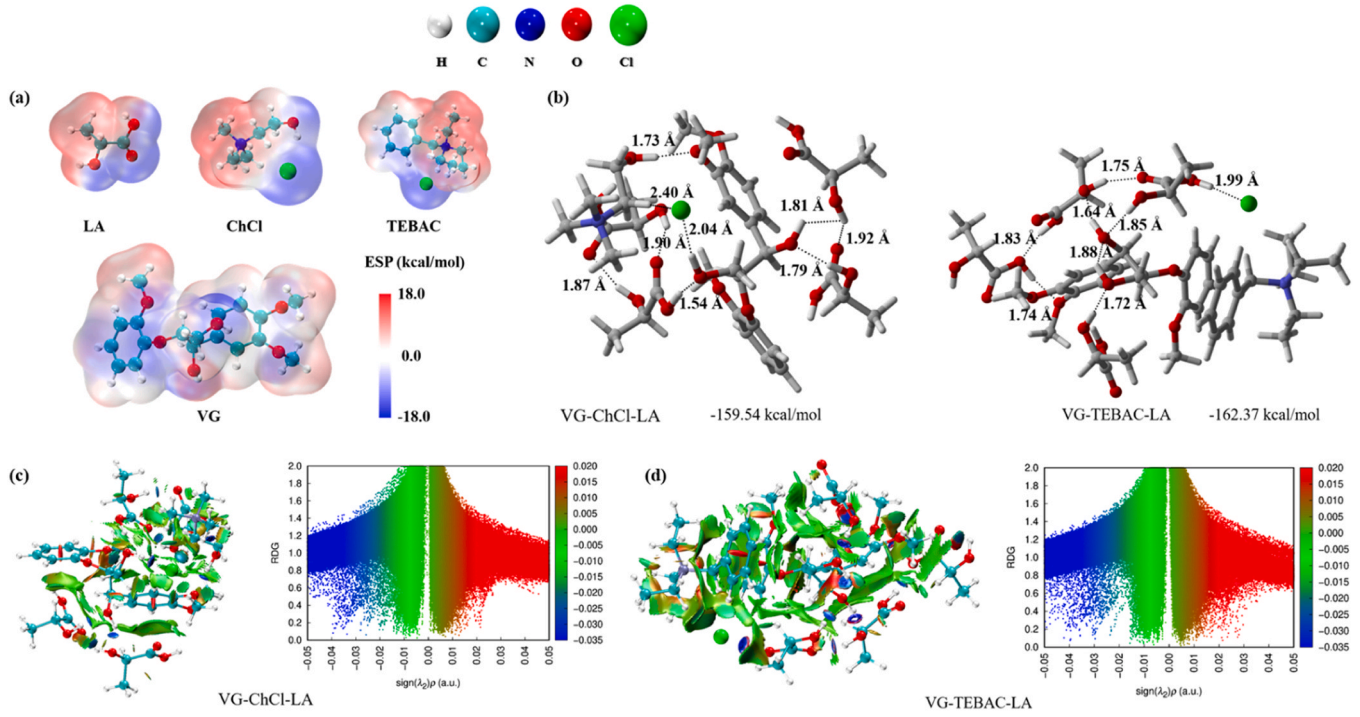


Fig. 4. DFT calculations of DES interactions with lignin. (a) The geometric configurations and electrostatic potential surfaces of LA, ChCl, TEBAC, and VG. (b) Stabilized structures of VG-ChCl-LA and VG-TEBAC-LA with the lowest energy. (c) RDG equivalent surface and scatter plot of the VG-ChCl-LA system. (d) RDG equivalent surface and scatter plot of the VG-TEBAC-LA system.

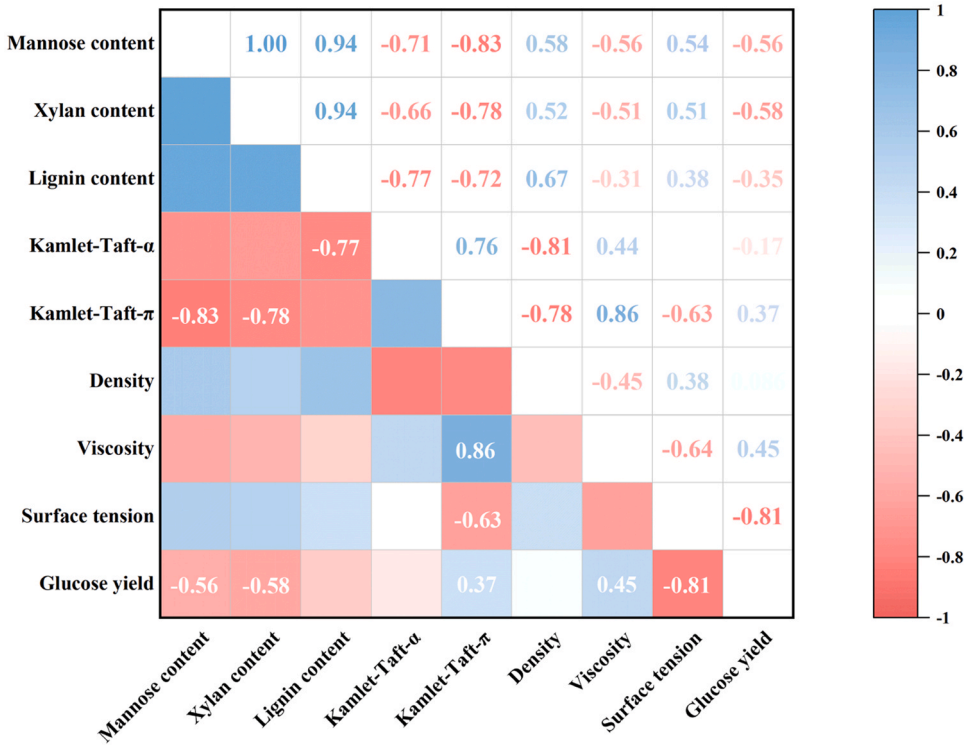


Fig. 5. Pearson correlation analysis of various physicochemical properties with enzymatic yield.

characteristics of DES influence the enzymatic hydrolysis of *Chinese fir* cellulose, a heatmap displaying Pearson correlation coefficients was created, as shown in Fig. 5. The results showed a less significant negative correlation between the glucose yield and the composition of the surface chemical fractions (correlation coefficients of −0.56, −0.58, and −0.35, respectively), which could be attributed to the selection of parallel pretreatment methods that did not form an effective gradient. The analysis showed that effectively removing hemicellulose and lignin improved the accessibility of polysaccharides, speeding up the enzymatic breakdown of cellulose and greatly increasing glucose production,

consistent with the results reported by Su et al. (2023). Additionally, the Kamlet-Taft π parameter and viscosity of DES positively correlated with glucose yield (correlation coefficients of 0.37 and 0.45 respectively), while surface tension showed a strong negative correlation (correlation coefficient of −0.81). This negative correlation was due to the reduction in surface tension and intermolecular cohesion under high temperatures, which facilitated the breakdown and reformation of hydrogen bond networks, increasing the likelihood of hydrogen bonding with lignocellulosic structures and aiding the removal of hemicellulose and lignin. Moreover, a high negative correlation existed between the Kamlet-Taft α parameter and lignin content (correlation coefficient of −0.77), confirming the link between hydrogen bond acidity (α) and the solvent's ability to extract lignin, consistent with the discussion in Section 3.1.3. Raising the Kamlet-Taft α value and the concentration of acid improved the ability of DES to donate protons, which promoted proton release and facilitated the breaking of chemical bonds in lignocellulose as well as the depolymerization of lignin (Kang et al., 2023). The analysis also revealed that the Kamlet-Taft π parameter correlated highly with the removal of hemicellulose components (correlation coefficients of −0.83 and −0.78), suggesting that DES with higher Kamlet-Taft π values should be chosen for effective hemicellulose removal. A comprehensive examination of the correlation factors underscored that the key variables influencing the preprocessing performance of DES solutions were the Kamlet-Taft α and π parameters, along with surface tension.

3.3. Optimisation of DES pretreatment biomass system

By analyzing the physicochemical properties of DES, along with theoretical calculations and practical applications, the combination of TEBAC/LA (1:4) was identified for its outstanding pretreatment effectiveness, and the best conditions for pretreatment were investigated. To examine how various temperatures impact the materials before and after pretreatment, an analysis was conducted on the components of *Chinese fir* residue that had been pretreated at different temperatures. The chemical composition and quality changes of the pretreated samples were shown in Table S4. With an increase in pretreatment temperature, there was a notable reduction in the content of hemicellulose sugars and lignin, which fell to 0.96 % and 1.88 %, respectively, at 150 °C. This indicated that high temperatures might cause the breakdown of chemical bonds among lignin, cellulose, mannan, and xylan, leading to the elimination of lignin (Qi et al., 2018). Simultaneously, the solid recovery rate also decreased, falling below 40 % when the pretreatment temperature hit 160 °C. Despite a reduction in mass during the pretreatment process, Fig. 6a showed that the retention rate of glucan remained high across all pretreatment temperatures, maintaining above 90 %. As the temperature increased, the corresponding yield of enzymatic hydrolysis glucose rapidly rose from 22.98 % to 97.09 % before slightly declining, demonstrating the outstanding pretreatment performance of TEBAC/LA (Liu et al., 2019, 2021). It was evident that for structurally *Chinese fir* residue, a significant increase in enzymatic hydrolysis yield was only achieved when lignin was substantially removed. In industrial production, an increase in temperature implies higher energy consumption. To ensure excellent pretreatment results while reducing energy consumption, the subsequent pretreatment temperature was set at 150 °C.

Table S5 displayed the chemical composition and quality changes of samples under different pretreatment durations. For the chemically resilient *Chinese fir* residue, increasing the pretreatment time was inevitable to achieve high enzymatic hydrolysis yields (Morais et al., 2020). As expected, the prolonged duration led to a significant reduction in the levels of hemicellulose sugars and lignin, which fell to 0.96 % and 1.88 % respectively after 90 min, indicating noticeable pretreatment effects. The substantial removal of hemicellulose and lignin contributed to high enzymatic hydrolysis yields. As shown in Fig. 6b, extending the pretreatment time from 15 min to 90 min resulted in an increase in glucose yield from enzymatic hydrolysis over 48 hours rose from 19.02 % to 97.09 %. This notable improvement also demonstrated the relationship between the yield of enzymatic hydrolysis and the removal of hemicellulose and lignin. When pretreatment durations were brief, the high content of lignin polymers, along with the coating effects of mannan and xylan in the cell wall, significantly obstructed enzymatic hydrolysis (Liu et al., 2021). As the pretreatment time was extended, the recovery rate of solids gradually decreased, dropping from 70.71 % to 41.58 % when the pretreatment duration was extended to 90 min. Although there was a mass loss during pretreatment, the retention rate of glucan remained high, consistently above 95 %. In industrial production, shorter pretreatment times implied higher production efficiency; therefore, an optimal pretreatment reaction time of 90 min was preferred.

3.4. Mass balance based on separation of three major components of TEBAC/LA pretreatment

TEBAC and LA were two readily available and non-hazardous reagents that could serve as ideal solvents for the economic and efficient

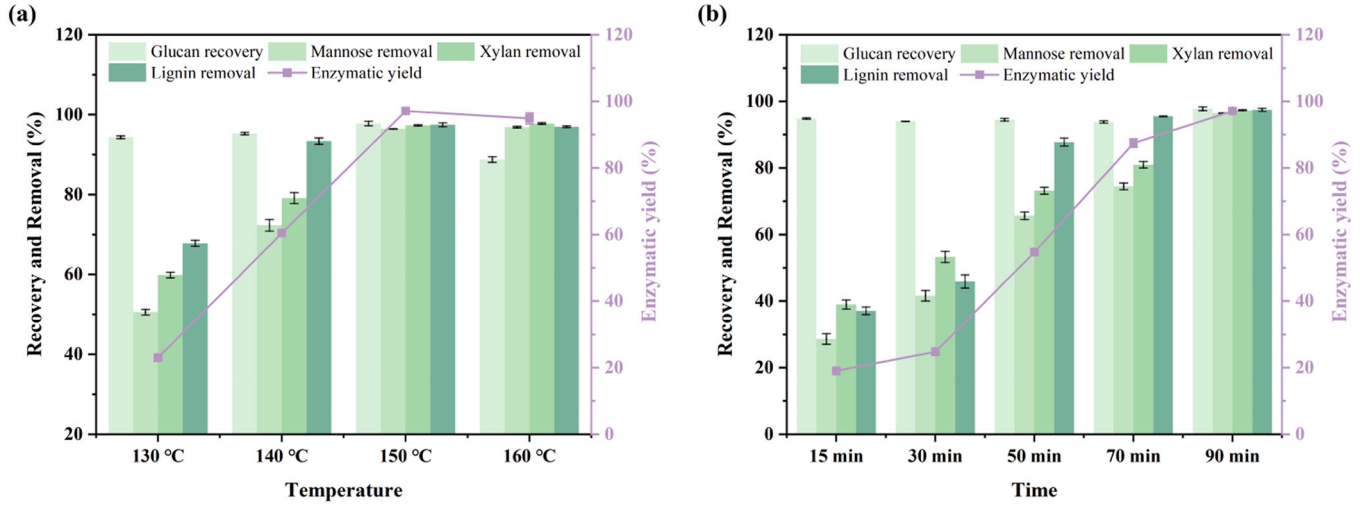


Fig. 6. Optimization of temperature and time for DES pretreated *Chinese fir* residue. (a) Retention, removal and 48 h enzymatic yield of each component after pretreatment of *Chinese fir* residue at different temperatures. (Reaction time: 90 min.) (b) Retention, removal and 48 h enzymatic yield of each component after pretreatment of *Chinese fir* residue at different times. (Reaction temperature: 150 ℃.).

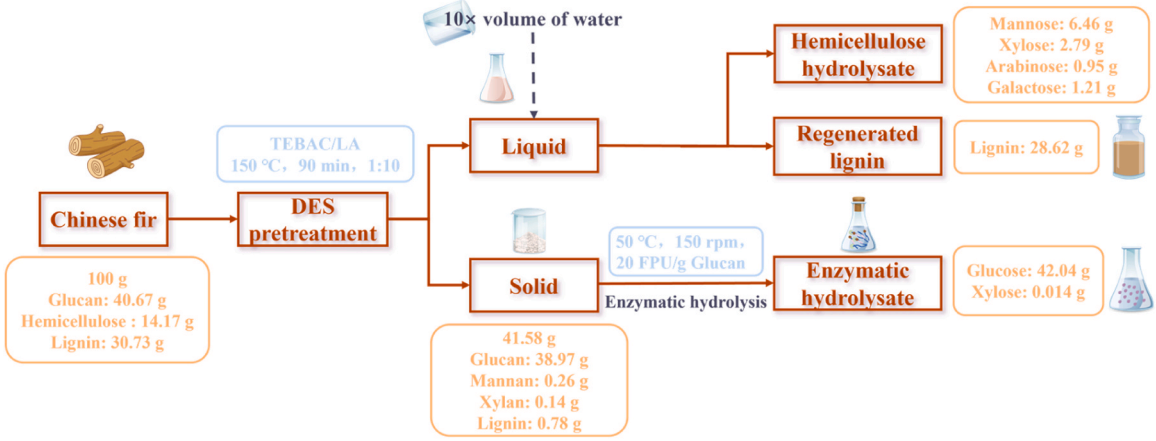


Fig. 7. Mass accounting for TEBAC/LA pretreated *Chinese fir* residue.

utilization of biomass. Liu et al. pretreated sugarcane bagasse (SCB) with TEBAC/LA, achieving an enzymatic yield of 88.23 % and lignin removal of 88.72 %. Even after five cycles, the enzymatic yield of SCB remained at 68.02 %, approximately 175 % higher than that of untreated SCB (Liu et al., 2021). In this study, the pretreatment of *Chinese fir* residue with TEBAC/LA aimed not only to provide a green and efficient treatment but also to enhance the value-added utilization of lignocellulose. Fig. 7 illustrated the entire process of pretreating *Chinese fir* residue under optimal conditions using TEBAC/LA. The mass balance calculations were based on the recovery rates and chemical compositions analyzed in Section 3.3.

Each 100.00 g of *Chinese fir* residue contained 40.67 g of glucan, 14.17 g of hemicellulose, and 30.73 g of lignin. Initially, the *Chinese fir* residue was pretreated with TEBAC/LA at a molar ratio of 1:4, conducted at a temperature of 150 °C for a duration of 90 min, utilizing a solid-to-liquid ratio of 1:10. After the reaction, solid-liquid separation was conducted. The recovered solid residue weighed 41.58 g, containing 38.97 g of glucan, 0.26 g of mannan, 0.14 g of xylan, and 0.78 g of lignin, achieving a glucan retention rate of 95.82 %. The DES pretreatment effectively separated the three major components, laying a solid foundation for subsequent enzymatic hydrolysis. The pretreated residue underwent enzymatic hydrolysis for 48 hours producing 42.04 g of glucose and 0.014 g of xylose. For the recovered liquid, hemicellulose soluble sugars and lignin were separated through precipitation with ten times the volume of water. Calculations showed that the recovered hemicellulose soluble sugars amounted to 11.41 g with a recovery rate of 80.52 %, and the regenerated lignin was 28.62 g with a recovery rate of 93.10 %. Overall, this study successfully achieved efficient separation of the three major lignocellulosic components using the TEBAC/LA DES system, providing a highly effective pathway for subsequent glucose fermentation and value-added lignin utilization, thereby opening new directions for the high-value utilization of biomass.

4. Conclusion

Effective pretreatment of lignocellulosic biomass waste is essential for enhancing subsequent enzymatic hydrolysis and enabling value-added utilization. By characterizing the physicochemical properties of a series of DES and conducting theoretical calculations, the pretreatment process of *Chinese fir* residue using TEBAC/LA was optimized to enhance enzymatic saccharification. Results indicated that under a TEBAC/LA (1:4) ratio, pretreatment at 150 °C for 90 min achieved a cellulose enzymatic hydrolysis rate of 97.09 % and a lignin removal rate of 97.41 %. Mass balance results showed that the hemicellulose recovery rate was 80.52 %, with 28.62 g of regenerated lignin and a recovery rate of 93.10 %. Molecular dynamics simulations demonstrated strong hydrogen bonding interactions between TEBAC/LA and lignin, promoting efficient lignin removal. Correlational factor analysis further revealed that high Kamlet-Taft α parameters, high Kamlet-Taft π parameters, and low surface tension significantly impact the pretreatment performance of DES. In summary, this study offers a viable pretreatment approach aimed at enhancing enzymatic saccharification and facilitating lignin recovery from lignocellulosic waste through rational design.

CRediT authorship contribution statement

Ji Chenjie: Writing – original draft, Investigation, Data curation. **Ouyang Jia:** Writing – review & editing, Funding acquisition, Conceptualization. **Weng Rou:** Methodology. **Yu Chang:** Validation. **Fan Yuyang:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.indcrop.2025.120779.

Data availability

Data will be made available on request.

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